An Analysis of Clinical Variables and Gene Expression Data across Biopsies from Patients with Lung Adenocarcinomata

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Introduction

Loading and Cleaning of Annotation and Expression Data

We want to clean up our annotation file. To do this we must first take a look at the file.

```
## # A tibble: 6 x 22
##   barcode   ajcc_pathologic_stage paper_patient paper_Sex paper_Age.at.diagnosis
##   <chr>     <chr>                 <chr>         <chr>     <chr>
## 1 TCGA-44-~ Stage IIB             TCGA-44-2665  FEMALE    55
## 2 TCGA-97-~ Stage IB              TCGA-97-7937  MALE      65
## 3 TCGA-44-~ Stage IIB             <NA>          <NA>      <NA>
## 4 TCGA-44-~ Stage IIB             TCGA-44-2665  FEMALE    55
## 5 TCGA-49-~ Stage IB              <NA>          <NA>      <NA>
## 6 TCGA-55-~ Stage IA              <NA>          <NA>      <NA>
## # i 17 more variables: paper_T.stage <chr>, paper_N.stage <chr>,
## #   paper_Tumor.stage <chr>, paper_Smoking.Status <chr>, paper_Survival <chr>,
## #   paper_Transversion.High.Low <chr>, paper_Nonsilent.Mutations <dbl>,
## #   paper_Nonsilent.Mutations.per.Mb <chr>,
## #   paper_Oncogene.Negative.or.Positive.Groups <chr>, paper_Fusions <chr>,
## #   paper_expression_subtype <chr>,
## #   paper_chromosome.affected.by.chromothripsis <chr>, ...
```

We can see here, that there are indeed values missing in a lot of the columns. Looking at the documentation for the data source ({{SOURCE}}), we can also see, that the first three segments = or the first 12 digits = of the barcode encode the patient ID. We thus replace paper_patient with the barcode trimmed to position 12.

Since we only want the barcode and paper columns, we discard anything not matching to that selection. This leaves us with 20 paper columns.

As this also makes the 'paper_' prefix superfluous we strip it from the paper columns. Furthermore, we noticed textual NAs, which were entered into the data set as '[unknown]', '[not available]' and '[Not Available]'. We thus strip these and replace them by real NAs. We perform the same operation on the expression data, whilst we're at it. The paper_Tumor.stage column encodes the Ajcc pathological stage. Thus, we replaced it with the ajcc staging.

When looking at the counts of NAs, the expression data are complete, whilst the annotation data have 5623 NAs in total distributed relatively evenly across all columns, with patient and Tumor.stage being the exceptions.

```
## [1] 0
```

```
## [1] 5347
```

```
##                                        column NA.count
## 1                                    patient        0
## 2                                        Sex      272
## 3                          Age.at.diagnosis      284
## 4                                    T.stage      275
## 5                                    N.stage      275
## 6                                Tumor.stage        0
## 7                             Smoking.Status      279
## 8                                   Survival      272
## 9                      Transversion.High.Low      271
## 10                       Nonsilent.Mutations      271
## 11                Nonsilent.Mutations.per.Mb      271
## 12       Oncogene.Negative.or.Positive.Groups      271
## 13                                   Fusions      444
## 14                        expression_subtype      271
## 15      chromosome.affected.by.chromothripsis      449
## 16                             iCluster.Group      271
## 17                CIMP.methylation.signature.      316
## 18 MTOR.mechanism.of.mTOR.pathway.activation      313
## 19                     Ploidy.ABSOLUTE.calls      271
## 20                     Purity.ABSOLUTE.calls      271
```

From the head-call above, we know that all of the text columns are characters, even when a factor or numeric type would be more appropriate. We thus go through the cleaned annotation table and factorise, where appropriate. Furthermore, where factorisation is inappropriate, we try to coerce each column to numeric type (accepting both comma and period as decimal delimiters) and discard all failed coercions. Lastly, we double check, that we see the intuitively appropriate classes for each column.

We know from the documentation of the data ({{SOURCE}}), that the leading numbers of the fourth segment tell us whether the sample is a tumour or not. We use this to encode a new column entitled 'is.tumour' storing that information. Since this segment only has the values of '01' and '11', and since '11' codes for healthy tissue and '01' for tumour tissue, a boolean suffices.

With our annotation data (and incidentally also expression data) cleaned up and usable, we move on to the next step.

Data Exploration

Clinical Annotation

Expression Data

Data Analysis

Dimensionality Reduction of Annotation Data

Differential Expression Analysis

Expression Correlation amongst Highly Differentially Expressed Genes

Predictive Modelling

Based on Expression Data

Target: Expression Subtype

First we initialise a mutable combined data set from our immutable annot and exp data sets, discard unnecessary annotation columns, and split the data into a training and testing data set. We also scale and centre both data sets based on the training data.

Using the scaled and centred training data, we then train randomForest and multinomial logistical regression models and make a prediction each.

We want to take a look at the 5 most important variables for both models. For the random forest regression, we simply use variable importance as the metric. For multinomial logistical regression, we use the euclidean deviance of the coefficients from zero for each gene. These differ, though there is overlap.

```
## [1] "FGA"    "PGC"    "FGL1"   "S100P"  "LGALS4"
```

```
## [1] "SCGB3A2"   "DSCAM.AS1" "FGL1"      "PGC"       "FGA"
```

We then walk through both lists of genes sorted by their importance to the models until we reach an overlaping selection of 5. The given depth shows us how many of the sorted list we need to include to get our overlap.

```
## $important.variables
## [1] "FGA"       "PGC"       "FGL1"      "FGB"       "DSCAM.AS1"
##
## $depth
## [1] 14
```

Target: Tumor Stage

We now do the exact same for the prediction of tumor stage, with the addition of splitting the tumour stages into only two categories: Early and Late.

If we look at the importance of the various genes to the prediction, we again see some overlap between the random forest and binomial logistical regressions in the top 5 most important variables.

```
## [1] "LGALS4" "EEF1A2" "MUC5AC" "PGC"    "BPIFA1"
```

```
## [1] "SFTPA2" "CLDN18" "MUC5B"  "BPIFA1" "CALCA"
```

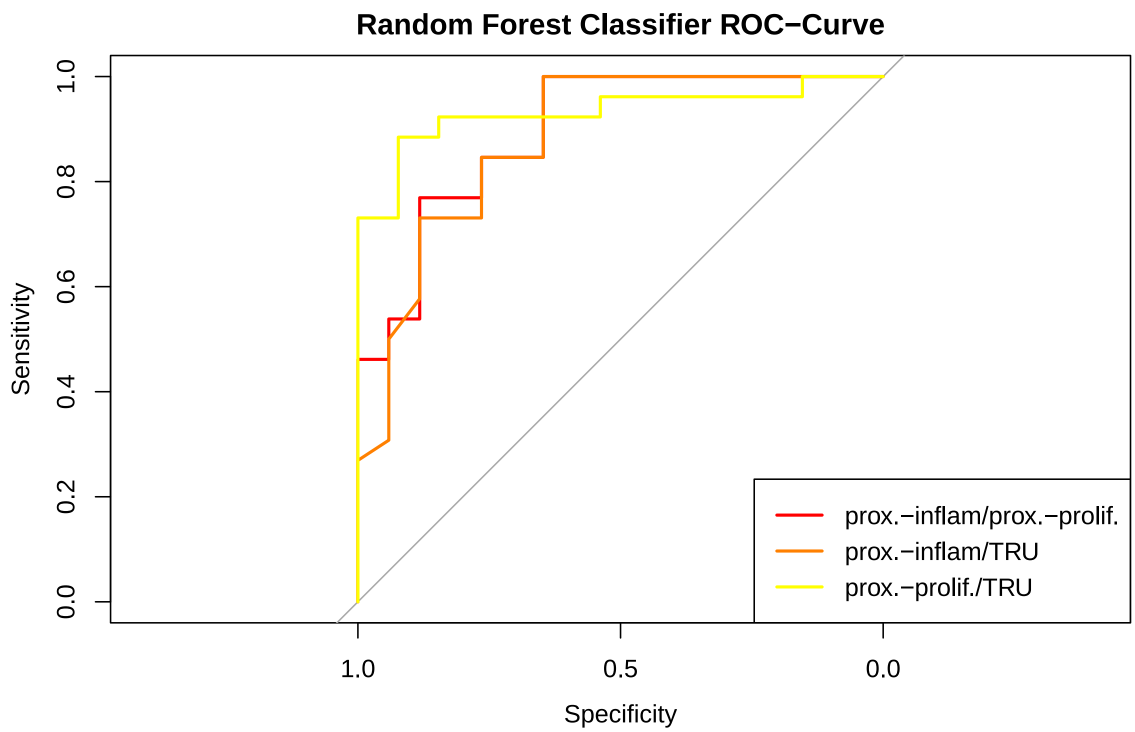
```
## [1] "AKR1B10" "AKR1C2"  "KLK12"   "SFTPC"   "EREG"
```

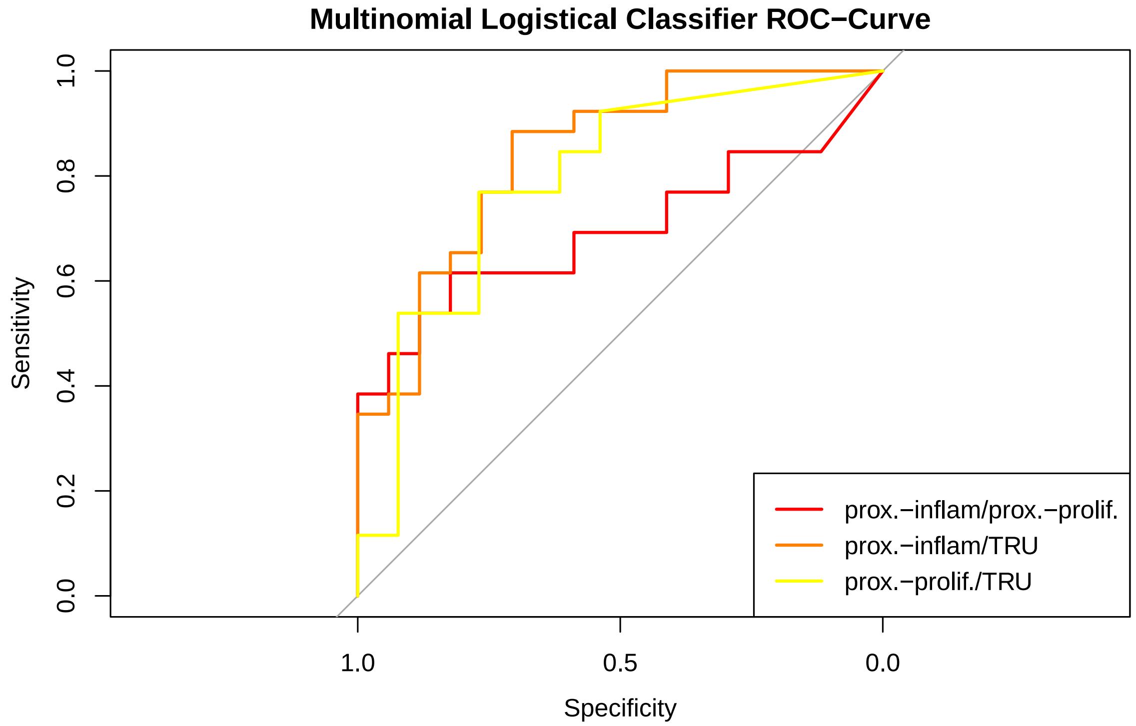
We thus again look at how far we need to go to find an overlap between all three models of 5 important variables.

```
## $important.variables
## [1] "BPIFA1"  "AKR1B10" "KLK12"   "CLDN18"  "SFTPA2"
##
## $depth
## [1] 8
```

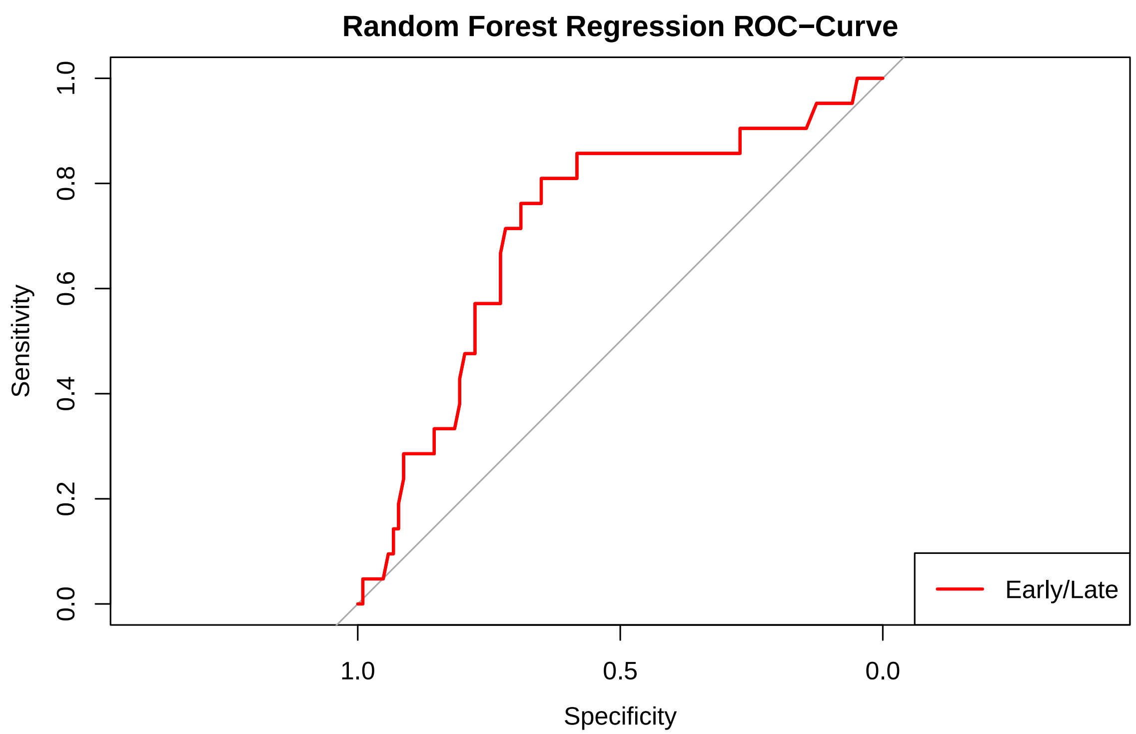
Quality and Comparison of Models

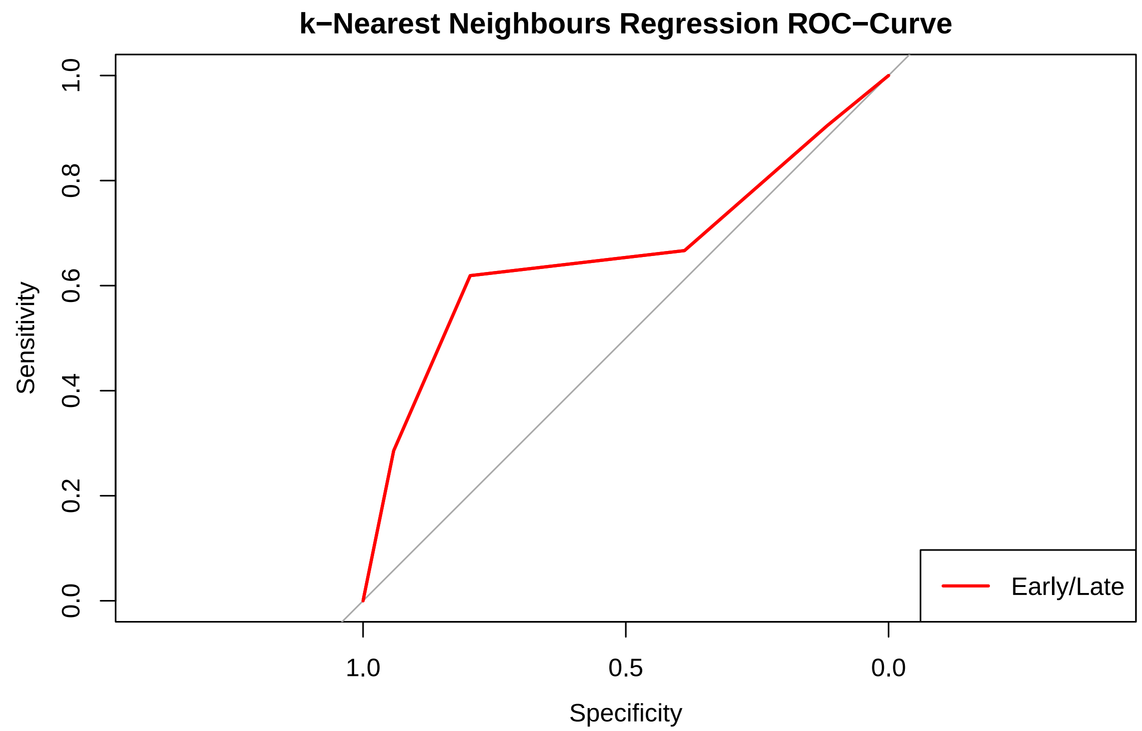
To compare the quality of the modesl, we take a look at ROC curves to ascertain the quality of the expression subtype classifiers. The random forest classifier outstrips the multinomial logistical one, especially when it comes to distinguishing proximal-inflammatory from proximal-proliferative biopsies.

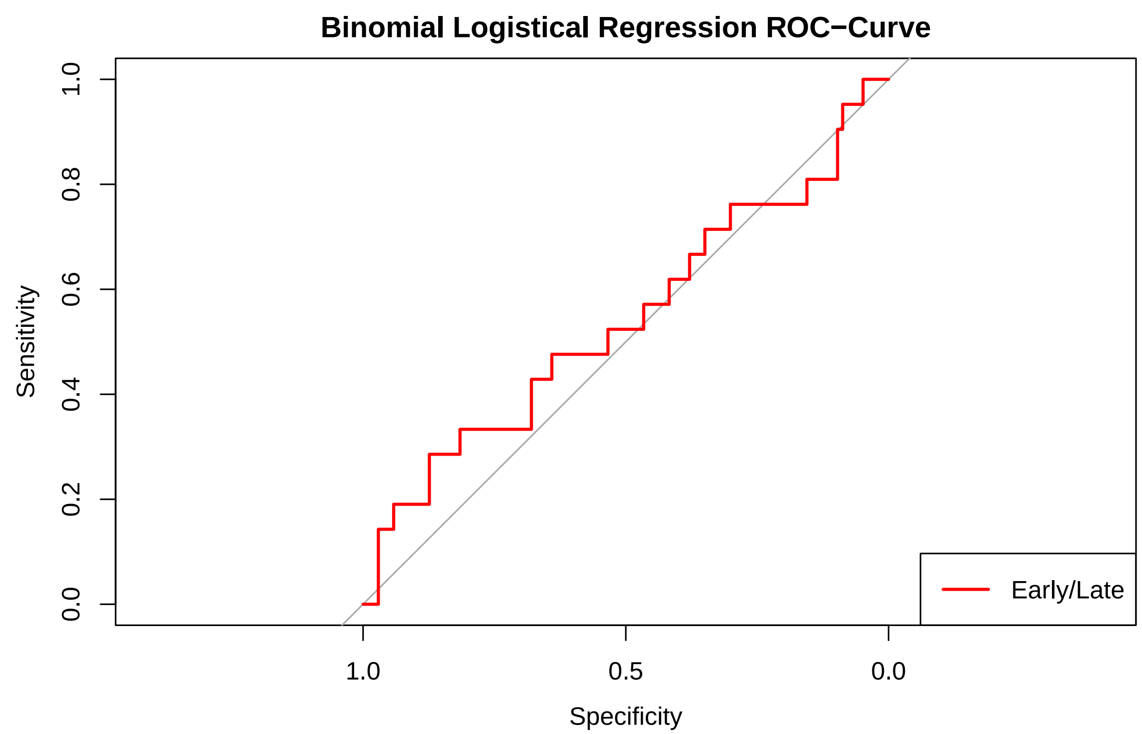




We now take a look at the ROC curves for the tumour stage classifiers. We see that random forest classification and kNN outstrip logistical regression.







When we look at key values for these classifiers (first table expression subtype classification, second table tumour stage classification) we can see why some of them don't perform well.

```
##                                     Random.Forest Multinomial.Logistical
## Accuracy                                0.7678571              0.6428571
## Macro-Averaged F1                       0.7401530              0.6172840
## Precision for Class: prox.-inflam       0.8181818              0.7000000
## Precision for Class: prox.-prolif.      0.6111111              0.4761905
```

```
## Precision for Class: TRU                0.8518519              0.7600000
## Recall for Class: prox.-inflam          0.5294118              0.4117647
## Recall for Class: prox.-prolif.         0.8461538              0.7692308
## Recall for Class: TRU                   0.8846154              0.7307692
```

```
##             Random.Forest Binomal.Logistical k.Nearest.Neighbours
## Accuracy        0.8225806          0.8306452            0.8306452
## F1              0.9026549          0.9074890            0.9066667
## Precision       0.8292683          0.8306452            0.8360656
## Recall          0.9902913          1.0000000            0.9902913
```

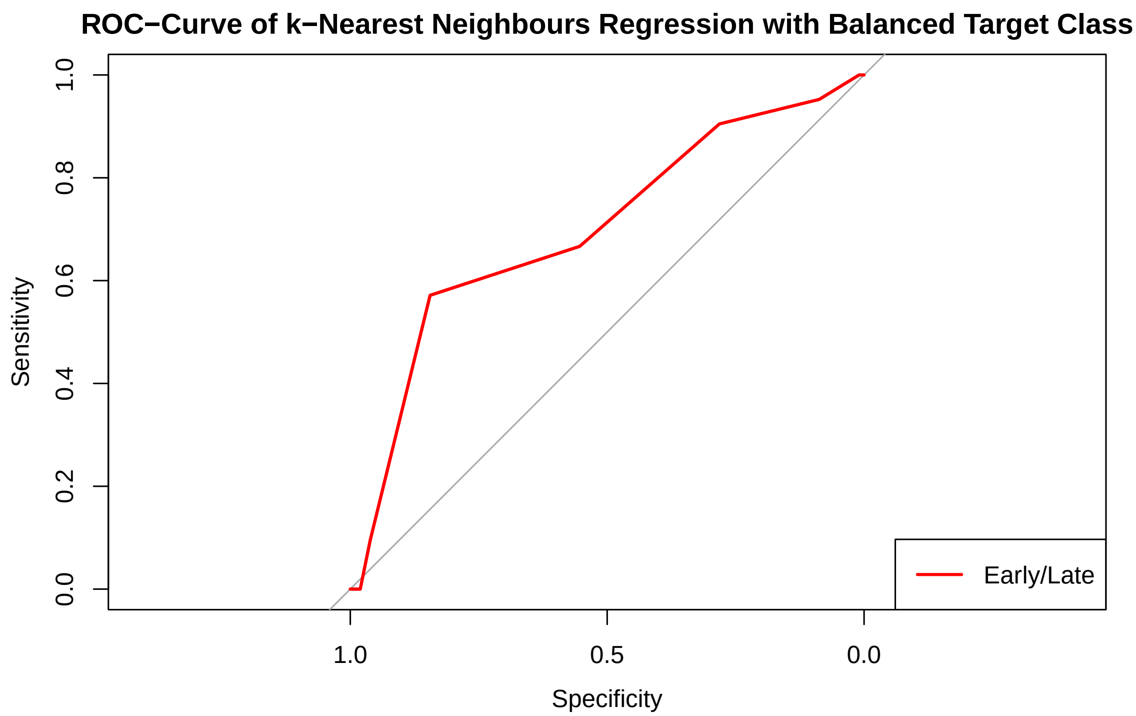
In the prediction of tumour stage, we have the problem, that our samples at Early stage far outnumber those at Late stage.

```
##
## Early  Late
##   223    64
```

Rerunning training for the best performer (highest accuracy and precision) using a down-sampled training data set worsens performance.

```
##  Accuracy        F1 Precision    Recall
## 0.5725806 0.6826347 0.8906250 0.5533981
```

Also the ROC Curve looks less healthy.



Based on Clinical Annotation

Integrative Model

Discussion

Conclusion